

Research Interests

Computational Social Science, Network Science and Statistical Network Models, LLM-Based Agentic Systems for Scientific Discovery, Reliability and Grounding of AI-Generated Research

Education

University of Southern California

Ph.D. Candidate, Industrial and Systems Engineering

M.S., Analytics

Los Angeles, CA, USA

Aug. 2023 – Expected May 2027

Aug. 2021 – Dec. 2022

Dissertation: Discovery of Network Formation Processes via LLM-Guided Exponential Random Graph Models

Advisor: Mayank Kejriwal

Key Courses: Design of Experiments and Quality Engineering, Stochastic Processes, Decision Analysis, Reinforcement Learning, Linear Optimization, Stochastic Optimization, Data Management, Data Mining, Text Analytics

The Chinese University of Hong Kong, Shenzhen

B.S., Statistics

Shenzhen, China

Sep. 2016 – May 2020

Key Courses: Probability and Statistics, Regression Analysis, Statistical Inference, Survey Sampling, Nonparametric Statistics, Categorical Data Analysis, Statistical Modelling in Financial Markets, Stochastic Calculus

Publications

Kejriwal, M., & Sun, Y. (2026). Building discoverative AI ecosystems for supporting high-quality agentic science. *AIAS+ 2026*. Forthcoming.

Sun, Y., & Kejriwal, M. (2026). Modeling complex associations between socio-demographic factors and climate policies using Conditional Inference Trees. *Humanities and Social Sciences Communications*.

Sun, Y., & Kejriwal, M. (2026). Structural modeling of campaign finance decisions in the U.S. House of Representatives. *PLOS Complex Systems*, 3(5), e0000104.

Sun, Y., He, D., & Kejriwal, M. (2026). Examining persistent inequities in financial complaint resolution for older Americans and veterans in the United States. *PNAS Nexus*, 5(4), pgag086.

Sun, Y., Shen, K., & Kejriwal, M. (2025). Backlash or reinforcement? Donald Trump's 2017 inauguration and shifting climate beliefs in the United States. *npj Climate Action*, 4, 26.

Sun, Y. (2025). Scalable estimation of exponential random graph models on large networks. *Companion Proceedings of the ACM Web Conference (TheWebConf)*, 721–724.

Gawin, C., Sun, Y., & Kejriwal, M. (2025). Navigating semantic relations: Challenges for language models in abstract common-sense reasoning. *Companion Proceedings of the ACM Web Conference (TheWebConf)*, 971–975.

Sun, Y., & Kejriwal, M. (2024). A study of firm-switching of inventors in Big Tech using public patent data. In *Social Network Analysis and Mining Applications in Healthcare and Anomaly Detection* (Ch. 12). Springer.

Sun, Y., & Kejriwal, M. (2023). A structural study of Big Tech firm-switching of inventors in the post-recession era. *Proceedings of the International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 670–677.

Sun, Y., & Kejriwal, M. (2023). DeepGraph: Multi-cluster interactive visualization of complex networks in a learned representation space. *Proceedings of the International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 427–430.

Under Review

Patil, V., Bacha, S. V., Yamani, R., Sun, Y., & Kejriwal, M. (2026). Beyond the star rating: LLM-assisted aspect analysis and large-scale text classification of 4.7 million restaurant reviews. Revised manuscript under review at *PLOS ONE*.

Sun, Y., & Kejriwal, M. (2026). Online sex-work advertising on Backpage in Southeast Asia, 2013–2018: A retrospective analysis of posting, place, and price. Revised manuscript under review at *SN Social Sciences*.

Sun, Y., & Kejriwal, M. (2026). FORGE: An agentic system for ERGM specification and interpretation in social networks. Submitted to *WSDM 2027*.

Sun, Y., et al. (2026). Computational social science for harm reduction: Privacy-preserving analysis of market structure in counter-trafficking work. Submitted to the *AAAI 2027 Artificial Intelligence for Social Impact Track*.

Under Preparation

Sun, Y., & Kejriwal, M. (2026). Hallucinated references in AI-generated publications: A cross-corpus audit. Manuscript in preparation.

Sun, Y., & Kejriwal, M. (2026). Interactive exploration of tie-formation hypotheses in complex networks. Manuscript in preparation (demonstration paper).

Sun, Y., & Kejriwal, M. (2026). Coordination and fragmentation in Latin American and Caribbean online sex work markets. Manuscript in preparation.

Sun, Y., & Kejriwal, M. (2026). A framework for organized control at the account level in online sex trafficking detection. Manuscript in preparation.

Research Funding

DARPA MAGICS, \$300,000. Primary proposal author; PI: Mayank Kejriwal. 2026.

Kroner Family Foundation. Philanthropic gift supporting research on human trafficking. 2024–2026.

Awards

Jenny Wang Excellence in Teaching Award (USC)	2024
ISE Department Outstanding Teaching Assistant of the Year (USC)	2024

Presentations

Poster Presentation, ‘Building reliable PhD-level agents for computational research’, Agentic AI Summit, University of California, Berkeley, Berkeley, CA, August 1–2, 2026.

Poster Presentation, ‘Mapping collaboration in AI-driven firms: A co-inventor dataset (2020–2025)’, International Conference of the Computational Social Science Society of the Americas (CSSSA), Washington, D.C., November 6–9, 2025.

Poster Presentation, ‘Reviewer rating variability, reviewer confidence, and language model-based sentiment prediction of machine learning conference submissions’, 10th International Congress on Peer Review and Scientific Publication, Chicago, September 3–5, 2025.

Oral Presentation, ‘Collaboration diversity and paper success: A network analysis of co-authorship in a major machine learning conference’, INSNA Sunbelt Conference, Paris, June 23–29, 2025.

Poster Presentation, ‘Scalable estimation of exponential random graph models on large networks’, The Web Conference (TheWebConf), May 13, 2025.

Oral Presentation, ‘Navigating semantic relations: Challenges for large language models in abstract common-sense reasoning’, The Web Conference (TheWebConf), May 13, 2025.

Presentations (continued)

Oral Presentation, ‘AI reveals systemic financial inequities: A study of CFPB complaints for older adults and veterans’, USC ShowCAIS Symposium, April 2025.

Oral Presentation, ‘Understanding the drivers of human trafficking using massive multi-partite network modeling and analytics’, USC Cloudwalkers Premiere, March 2025.

Oral Presentation, ‘A structural study of Big Tech firm-switching of inventors in the post-recession era’, Advances in Social Networks Analysis and Mining (ASONAM), November 7, 2023.

Oral Presentation, ‘DeepGraph: Multi-cluster interactive visualization of complex networks in a learned representation space’, Advances in Social Networks Analysis and Mining (ASONAM), November 6, 2023.

Poster Presentation, ‘Modelling campaign finance data using network science to map structures and flows of influence’, Southern California Applied Mathematics Symposium (SOCAMS), April 22, 2023.

Media Coverage

Public Writing:
‘5 lessons from 20 years of climate surveys: Political party matters, but it doesn’t explain everything’, *The Conversation*, July 21, 2026. Syndicated by *The Washington Post*, *NationofChange*, and *ArcaMax*.

Research Coverage:
‘A majority of Americans don’t actually fit either climate camp’, *Anthropocene Magazine*, August 4, 2026.

‘Financial complaint delays hit seniors and veterans hardest, with gaps widening over time’, *Phys.org*, April 16, 2026.

‘Older Americans and veterans wait longer for financial justice’, *USC Information Sciences Institute*, April 16, 2026.

‘Big Tech inventor mobility leads to higher productivity’, *USC Information Sciences Institute*, November 7, 2023.

Public Scholarship

Global Trafficking Intelligence Hub 2024 – Present

Founder and Lead Developer, globaltrafficking.org

Built and maintain a public knowledge hub on human trafficking, hosted by the USC Artificial Intelligence and Complex Systems group with support from the Kroner Family Foundation. Curates international reports, country- and region-level assessments, legal frameworks, datasets, and peer-reviewed research across eight world regions, with source-dated entries and a searchable research interface for practitioners, policymakers, and researchers.

Teaching Experience

University of Southern California (Teaching Assistant)

ISE 547: Applied Generative Artificial Intelligence for Enterprises	Fall 2025
ISE 529: Predictive Analytics	Fall 2025
ISE 561: Economic Analysis of Engineering Projects	Summer 2025
ISE 540: Text Analytics	Fall 2023, Spring 2025, Summer 2025

Mentoring

Shama Maganur (Montana State University)	Spring 2026
Richard Joungyoon Kim (CURVE Program; USC)	Fall 2024 – 2025
Angel Otters (Directed Research; USC)	Fall 2024
Daiqi He (Directed Research; USC)	Summer 2024 – 2025
Erica Okeh (Viterbi SURE Program; Howard University)	Summer 2024
Herry Wang (Directed Research; USC)	Summer 2024
Shree Vaishnavi Bacha (Directed Research; USC)	Spring 2024 – 2025
Vishal Patil (Directed Research; USC)	Spring 2024 – 2025
Yining Ma (Directed Research; USC)	Spring 2024
Revanth Yamani (Directed Research; USC)	Spring 2024
John Lim (REU Summer Program; Northwestern University)	Summer 2023

Professional Service

Journal Reviewer for:

Social Network Analysis and Mining (Springer)

Frontiers in Political Science

Frontiers in Sociology

Conference Reviewer for:

Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026

Research Experience

GRAIL

Research Scientist

Los Angeles, CA

May 2026 – Aug. 2026

Built LLM-based agentic systems for scientific research workflows, including an autonomous research pipeline that decomposes questions into executable steps and orchestrates tool calls across retrieval, analysis, and code execution; a retrieval-augmented grant recommender; and a hallucination checker that audits generated scientific text against source documents and bibliographic databases, with an accompanying evaluation harness and error taxonomy.

USC Information Sciences Institute

Research Engineer

Marina del Rey, CA

Jan. 2023 – Aug. 2023

Conducted applied research in the Artificial Intelligence and Complex Systems group; designed and ran controlled experiments, implemented model variants and evaluation pipelines, and contributed to academic publications through manuscript drafting and preparation of experimental results.

Datago Technology Limited

Quantitative Researcher Intern

Hong Kong

May 2021 – Sep. 2021

Built data pipelines to extract and standardize text features for sentiment-based factors; implemented factor construction logic from raw forum data and ran signal diagnostics and walk-forward backtests.

Technical Skills

Methods: Exponential Random Graph Models, Statistical Network Inference, Community Detection & Centrality Analysis, Graph Embeddings, Supervised & Unsupervised Learning, Text Classification & Topic Modeling

LLM & Agentic Systems: Agent Orchestration, Retrieval-Augmented Generation, Tool Use & Function Calling, LLM Evaluation Pipelines

Software: Python (PyTorch, Transformers, scikit-learn, pandas), R (ergm, statnet), SQL, Neo4j, Gephi